Tyler Jay Perkins

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Professional Experience

Teaching Assistant - Computer Science

Utah Valley University, Orem, UT *Aug 2026 - Present*

Courses: C Programming, C++ Software Dev, Java Software Dev, Database Theory, Analysis of Programming Languages

- Grade, tutor, and teach students across five undergraduate CS courses, one-on-one in person and remotely over Teams
- Compile, run, and test every submission against the spec, returning the student's own code corrected to show what failed
- Coordinate with multiple professors and TAs to keep grading consistent across sections and flag commonly missed requirements

Freelance Full-Stack Developer

Client: Honey Beehavior Analysis, Remote *Jan 2026 - Present*

Project Stack: React, Vite, Tailwind CSS, NestJS, TypeScript, PostgreSQL (Neon), Microsoft Entra ID (MSAL), Jest, Vitest

- Built a role-based portal for an ABA clinic where RBTs submit requests, daily checklists, and receipt-backed reimbursements by phone
- Designed a layered NestJS API on PostgreSQL; each status change is one guarded SQL update with its audit row, so reviews never collide
- Synced every signed-in screen live over Server-Sent Events, with optimistic updates and an offline queue that replays checklist work
- Secured sign-in with Microsoft Entra ID and role-based access, backed by 87 automated tests including concurrency and permission tests

Software Engineer Summer Internship - Full Stack Developer

Treehouse Strategy & Communication, Remote *Jul 2021 - Oct 2021*

Project Stack: React, ASP.NET, Web API, .NET Framework 4.7.1, Entity Framework, Git

- Developed and implemented a login system using React, ASP .NET, Web API, .NET Framework 4.7.1, and Entity Framework
- Collaborated with multiple development teams using Git for version control and conducted code reviews with Senior developers
- Participated in Agile development sprints, attending daily Scrums and contributing to Sprint planning

Selected Projects

More projects at TylerJayP.com and github.com/TylerJayP

Lockbox - Encrypted Desktop Password Manager

Project Stack: C#, .NET 10, Avalonia (MVVM), libsodium (NSec), Velopack, xUnit, GitHub Actions

Repository: github.com/TylerJayP/Lockbox (public, with signed Windows releases)

- Built a local-first password manager for logins, cards, Wi-Fi networks, and notes, kept in one encrypted vault file with no server or account
- Required a master password plus a key file to unlock, with Argon2id key stretching and XChaCha20-Poly1305 encryption from libsodium
- Made saves crash-safe and tamper-evident: each save is decrypted back before an atomic swap, and rollbacks to older copies are flagged
- Shipped signed auto-updates through Velopack and GitHub Releases, verifying an Ed25519 signature and SHA-256 hash before installing
- Covered crypto, the vault, import, export, and update signing with 98 xUnit tests, plus a headless end-to-end UI check in GitHub Actions

4380 Microprocessor Emulator & Assembler

Project Stack: C++, Python, GoogleTest, CMake

Repository: UVU Policy requires repository to be private. Available upon request

- Developed a complete microprocessor emulator simulating a custom 32-bit architecture with 22 registers, byte-addressable memory, and little-endian instruction encoding
- Implemented fetch-decode-execute cycle supporting 40 operations across instruction types including jumps, arithmetic, memory operations, and system traps
- Built a two-pass assembler in Python translating custom 4380 assembly language into 4380 machine code, resolving labels and encoding all 40 supported instructions
- Implemented error handling for invalid instructions, memory bounds checking, and division-by-zero protection
- Created a unit test suite using GoogleTest to validate instruction execution, register operations, and memory management

Parallel Dense & Sparse Matrix Multiplication on the Bridges-2 Supercomputer

Project Stack: C, MPI (Open MPI), SLURM, Linux, Bridges-2 (Pittsburgh Supercomputing Center)

Repository: Available upon request

- Implemented sequential, row-wise, and column-wise decomposed matrix multiplication in C with MPI, run as SLURM batch jobs on Bridges-2
- Extended to sparse matrices (70–90% zeros), reordering the loops to i-k-j so each zero in A skips its full row of multiply-adds
- Handled uneven splits with MPI_Scatterv/MPI_Gatherv, compared balanced vs. unbalanced remainder policies, and measured per-rank load imbalance (max / average compute time) caused by irregular non-zero distribution
- Verified both parallel versions bit-identical to the sequential baseline across 222 checks with up to 16 processes
- Instrumented phase timing with MPI_Wtime and MPI_Reduce, benchmarked 21 configurations, and traced a 5.9–7.1× column-wise speedup at large sizes to cache behavior on AMD EPYC 7742 nodes

Education

Master of Science Expected Fall 2028

Currently attending the MCS program at Utah Valley University

Bachelor of Science - Computer Science Apr. 2026

Utah Valley University, Orem, UT

Technical Skills

Proficient: C/C++, C#, Python, Java, JavaScript, MySQL

Embedded & Parallel: STM32, MPI, SLURM

Tools & Frameworks: Git, CMake, GoogleTest, xUnit, React, ASP.NET, Avalonia, Entity Framework, NestJS, PostgreSQL, AWS Amplify, Azure, Anthropic API